

MoneyPrinter

Autonomous Algorithmic Trading Bot — Simple Overview

What Is It?

MoneyPrinter is a fully automated stock trading bot that runs on GitHub Actions at no server cost. It scans a universe of ~150 large/mid-cap stocks every trading day, scores them with 30+ technical indicators, optionally validates high-confidence picks with Claude AI, and places bracket orders on Alpaca Paper Trading. Everything is free by default — no live money, no server.

How a Trade Happens (Step by Step)

Step	When	What Happens
1. Discovery	08:30 AM ET	Scans ~150 stocks for unusual volume or big price moves. Top 10 movers saved to discovered_tickers.js
2. Pre-Market	09:00 AM ET	Loads the discovered tickers and checks news headlines + gap analysis.
3. Score	9:30 AM+	Each ticker runs through 30+ technical indicators across 5 categories (Trend, Momentum, Volatility, Volume, News)
4. Gate Checks	9:30 AM+	Signal must pass: kill-switch check, max-trades cap, stop-width limit, regime filter, sector cap, portfolio exposure
5. Claude AI	9:30 AM+	If USE_CLAUDE=true, Claude reviews every signal and provides entry price, stop loss, take profit, and a confidence score
6. Order	9:30 AM+	A bracket order is placed on Alpaca: limit entry + stop loss + take profit. Position size is 2% of portfolio, scaled by confidence
7. Monitor	Ongoing	Open positions are checked every scan cycle for stop/target breach, time-based exits, and trailing stops.
8. EOD Summary	4:15 PM ET	Closed trades are reconciled, P&L logged to the database, and a daily summary written.

Scoring Engine

Every ticker earns bull (+) and bear (–) points. **Net Score = Bull – Bear**. **Confidence = Net ÷ 150** (capped at 1.0).

Category	Indicators
Trend	EMA 9/21/50/200 alignment, ADX, MACD, Parabolic SAR, VWAP
Momentum	RSI, Stochastic RSI, CCI, Williams %R
Volatility	Bollinger Bands (%B), Keltner Channels, ATR, BB squeeze
Volume	OBV, Volume Ratio, MFI
News	yfinance headlines, TextBlob sentiment, hype/boost keywords
Fundamentals	Revenue growth, EPS beats, institutional ownership, short interest

Trade signal thresholds:

- BUY: Net ≥ 65 AND Confidence ≥ 0.47
- SHORT: Net ≥ 70 AND Confidence ≥ 0.70
- HOLD: Everything else

Risk Management

Rule	Value
Base risk per trade	2% of portfolio
Max position size	10% of portfolio
Max total exposure	60% of portfolio
Max new entries per session	2
Max sector positions	2 per sector
Stop loss	1.5× ATR below entry
Take profit	3.75× ATR above entry (~2.5× R/R)
Kill switch	Halt all orders if daily P&L < −3%
Stop width cap	Block order if stop > 8% from entry
VIX scaling	Position size reduced as VIX rises; new longs halted above VIX 35

Strategies

Strategy	Trigger	Hold Period
trend_follow	Full EMA stack aligned + ADX > 22 + MACD rising	2–10 days
breakout	Price breaks R1 or 52-week high on volume	2–10 days
squeeze_breakout	BB squeeze resolves + Keltner breakout	2–10 days
breakdown	Price breaks S1 or 52-week low (short)	2–10 days
mean_reversion	RSI/BB/CCI deeply oversold — bounce expected	Same day to 2 days
news_momentum	Strong news catalyst + EMA alignment or volume surge	Same day to 2 days

Daily Schedule (GitHub Actions)

Workflow	Time (EDT)	What It Does
discovery.yml	8:30 AM	Screens ~150 stocks for active movers, saves top 10
premarket.yml	9:00 AM	News and gap analysis on discovered tickers
trading_day.yml	9:30 AM	Continuous scan + score + execute loop all day
eod_summary.yml	4:15 PM	Reconcile positions, log daily P&L, write summary
test_ai_filter.yml	Manual	Test Claude filter on any tickers via workflow_dispatch

Workflows run on GitHub's free tier — no server or cloud account needed. The bot determines if the market is open by querying the Alpaca calendar API, so it automatically handles holidays, early closes, and edge cases like Juneteenth.

Data Flow

Inputs:

- Alpaca: live prices, portfolio positions, order placement
- yfinance: 2 years of OHLCV bars for indicator calculation

- yfinance / NewsAPI: latest headlines for sentiment scoring

Outputs:

- data/live_feed.json — decision feed for the Vercel dashboard (updated every scan)
- data/trades.db — SQLite trade log (or Turso if TURSO_DATABASE_URL is set)
- GitHub Actions logs — full indicator + reasoning trace per scan
- discovered_tickers.json — top movers from each discovery session

Live Dashboard (Vercel)

A single-file HTML dashboard lives in vercel-dashboard/index.html. Deploy it to Vercel for free in ~2 minutes (connect repo → set Root Directory to vercel-dashboard → Deploy). It auto-updates every 30 seconds by reading live_feed.json from GitHub.

Three views:

- Portfolio — open positions with unrealized P&L; and hold time
- Trades — closed trade history with entry/exit prices and realized P&L;
- Bot Status — real-time decision feed for every ticker scanned

Setup in 5 Minutes

1. Fork or clone the repo to your GitHub account.
2. Go to Settings → Secrets → Actions and add:

Secret	Where to Get It
ALPACA_API_KEY	alpaca.markets → Paper Trading → API Keys
ALPACA_SECRET_KEY	Same page
ANTHROPIC_API_KEY	console.anthropic.com (optional — only needed if USE_CLAUDE=true)
NEWS_API_KEY	newsapi.org — free tier, optional
TURSO_DATABASE_URL	turso.tech — optional, for persistent remote trade history
TURSO_AUTH_TOKEN	Same as above

3. Enable write permissions: Settings → Actions → General → Workflow permissions → Read and write.
4. Push any change to main — the workflows will start running automatically on the next market day.

Cost

Service	Cost
Alpaca Paper Trading	Free
GitHub Actions	Free (public repos)
yfinance data	Free
Vercel dashboard	Free
NewsAPI (optional)	Free (100 req/day)
Claude AI (optional)	~\$0.01–0.05 per scan if USE_CLAUDE=true

Service	Cost
Turso DB (optional)	Free tier covers trading volume easily
Total	~\$0 to \$1/month

Disclaimer: This bot trades on Alpaca Paper Trading by default — no real money. It is provided for educational and research purposes only and is not financial advice. Never trade real money without fully understanding the risks.